Unit 5: Periodic Functions

Honors Precalculus

Student

Contents of this unit:

Contents

5.1 Graphs of Sine and Cosine

Objectives

- Sketch the parent graphs of $y = \sin x$ and $y = \cos x$.
- Determine the amplitude, period, and midline of sinusoidal graphs.
- Identify the parameters a, b, c, d in general sinusoidal forms.
- Write a trigonometric equation from a given graph.

Periodic Functions

A function f is **periodic** if there exists a positive constant P such that

$$f(x + P) = f(x)$$

for all x in the domain of f. The smallest such positive number P is called the ________ of the function.

The ________ ________ of a periodic function is the horizontal line halfway between its maximum and minimum values:

$$y = \_\_\_\_\_\_\_\_$$

The ________ is half the distance between the maximum and minimum values:

$$\text{amplitude} = \_\_\_\_\_\_\_\_$$

Parent Graphs: $y = \sin x$ **and** $y = \cos x$

The Sine Function

x	$y = \sin x$
0	0
$\pi/4$	$\sqrt{2}/2 \approx 0.7$
$\pi/2$	1
$3\pi/4$	$\sqrt{2}/2 \approx 0.7$
π	0
$5\pi/4$	$-\sqrt{2}/2 \approx -0.7$
$3\pi/2$	-1
$7\pi/4$	$-\sqrt{2}/2 \approx -0.7$
2π	0

The Cosine Function

x	$y = \cos x$
0	1
$\pi/4$	$\sqrt{2}/2 \approx 0.7$
$\pi/2$	0
$3\pi/4$	$-\sqrt{2}/2 \approx -0.7$
π	-1
$5\pi/4$	$-\sqrt{2}/2 \approx -0.7$
$3\pi/2$	0
$7\pi/4$	$\sqrt{2}/2 \approx 0.7$
2π	1

Fact

Characteristics of Parent Sine and Cosine Functions:

- **Domain:** ____________ **Range:** ____________
- **Period:** ________ **Amplitude:** ________ **Midline:** ________
- **y-intercept:** Sine: ________ Cosine: ________
- **x-intercepts:** Sine: ________ Cosine: ________ (where $n \in \mathbb{Z}$)
- **Parity / Symmetry:**
  - Sine is ________ (symmetric about the ________): $\sin(-x) = \_\_\_\_\_\_\_\_$
  - Cosine is ________ (symmetric about the ________ ________): $\cos(-x) = \_\_\_\_\_\_\_\_$

General Sinusoidal Forms

General Transformations

Any transformed sine or cosine function can be modeled by:

$$y = a\sin\big(b(x-c)\big) + d \qquad \text{or} \qquad y = a\cos\big(b(x-c)\big) + d$$

where each parameter controls a specific geometric transformation:

- **Amplitude:** ________. If $a < 0$, the graph is ________ ________ across the midline.
- **Frequency Coefficient:** b affects the horizontal scale. The **Period** P is given by:

$$P = \underline{\qquad\qquad}$$

- **Phase Shift:** ________ (horizontal shift). Positive c shifts ________, negative c shifts ________.
- **Vertical Translation:** ________. The horizontal line ________ becomes the new midline.

Note

The Factoring Warning: Always factor out the coefficient b from the inner expression to find the correct phase shift:

$$bx - c = b\left(x - \frac{c}{b}\right)$$

If written as $y = \sin(bx - c)$, the phase shift is $\frac{c}{b}$, not c. Always factor first!

Examples

Example 1

Find the amplitude, period, phase shift, and vertical translation of:

$$y = 3\sin\left(\frac{x}{2}\right)$$

Example 2

Find the amplitude, period, phase shift, and vertical translation of:

$$y = \frac{1}{2}\sin\left(x - \frac{\pi}{3}\right)$$

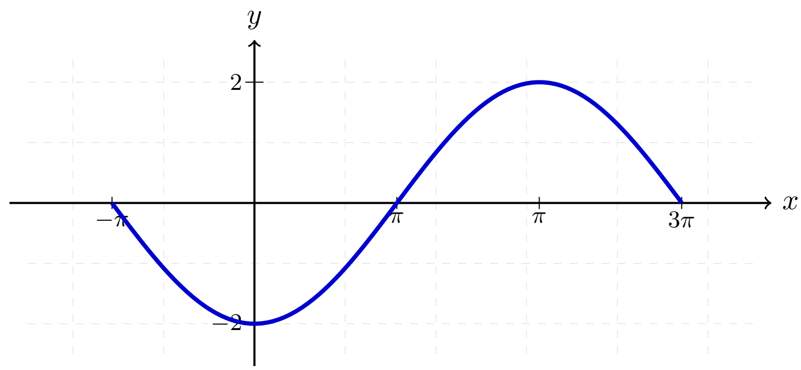
Example 3

Find the amplitude, period, phase shift, and vertical translation of:

$$y = -3\cos(2\pi x + 4\pi)$$

Example 4

Find a sinusoidal equation of the form $y = a\sin(b(x-c))$ for the graph below.



5.2 Graphs of Sine and Cosine (Part 2)

Objectives

- Identify the 5 key points (maxima, minima, and midline intercepts) of parent sine and cosine functions.
- Apply a systematic 5-step graphing procedure using t-tables to construct transformed graphs.
- Sketch transformed sine and cosine graphs including reflections, horizontal/vertical stretches/compressions, and horizontal/vertical translations.

The 5 Key Points

Every cycle of a sinusoidal graph is defined by **5 key points** that divide the period into 4 equal quarters:

- **Sine parent key points:**

$$(0, \underline{\hspace{2cm}}) \to \left(\frac{\pi}{2}, \underline{\hspace{2cm}}\right) \to (\pi, \underline{\hspace{2cm}}) \to \left(\frac{3\pi}{2}, \underline{\hspace{2cm}}\right) \to (2\pi, \underline{\hspace{2cm}})$$

  Pattern: $\underline{\hspace{2cm}}\ \underline{\hspace{2cm}} \to \underline{\hspace{2cm}} \to \underline{\hspace{2cm}}\ \underline{\hspace{2cm}} \to \underline{\hspace{2cm}} \to \underline{\hspace{2cm}}\ \underline{\hspace{2cm}}$

- **Cosine parent key points:**

$$(0, \underline{\hspace{2cm}}) \to \left(\frac{\pi}{2}, \underline{\hspace{2cm}}\right) \to (\pi, \underline{\hspace{2cm}}) \to \left(\frac{3\pi}{2}, \underline{\hspace{2cm}}\right) \to (2\pi, \underline{\hspace{2cm}})$$

  Pattern: $\underline{\hspace{2cm}} \to \underline{\hspace{2cm}}\ \underline{\hspace{2cm}} \to \underline{\hspace{2cm}} \to \underline{\hspace{2cm}}\ \underline{\hspace{2cm}} \to \underline{\hspace{2cm}}$

The 5-Step Graphing Procedure

Fact

To graph a transformed function of the form $y = a \cdot f(b(x - c)) + d$ (where f is sin or cos):

1. **T-Table:** Write the parent function's t-table containing the 5 key points.
2. **Analyze:** Determine the transformations:
   - Horizontal (affecting x): Period $P = \frac{2\pi}{|b|}$ and Phase Shift c.
   - Vertical (affecting y): Amplitude $|a|$, reflection (if $a < 0$), and Vertical Shift d.
3. **New x-coordinates:** Convert parent x coordinates using:

$$x_{\text{new}} = \underline{\hspace{2cm}}$$

4. **New y-coordinates:** Convert parent y coordinates using:

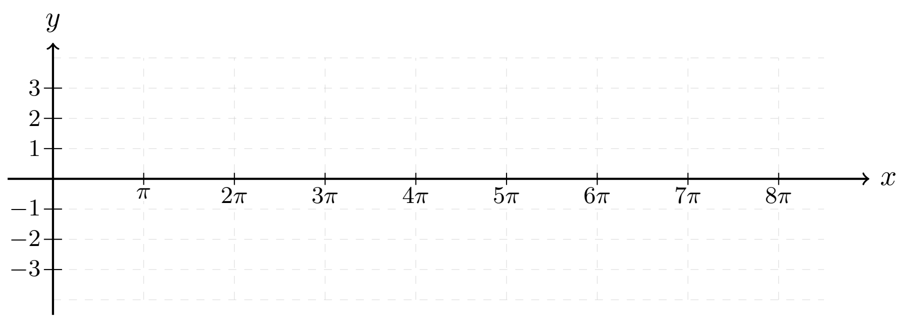
$$y_{\text{new}} = \underline{\hspace{2cm}}$$

5. **Plot and Sketch:** Set up a scaled grid, plot the 5 points, and draw a smooth wave. Extend the wave to multiple periods by repeating the pattern.

Examples

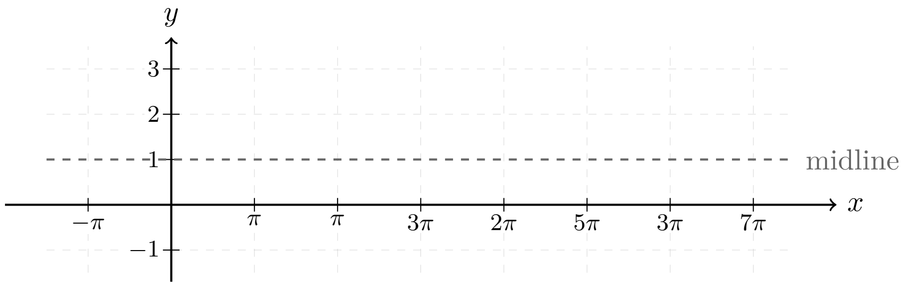
Example 1

Sketch the graph of $y = 3\sin\left(\frac{x}{2}\right)$ over two full periods starting at $x = 0$.



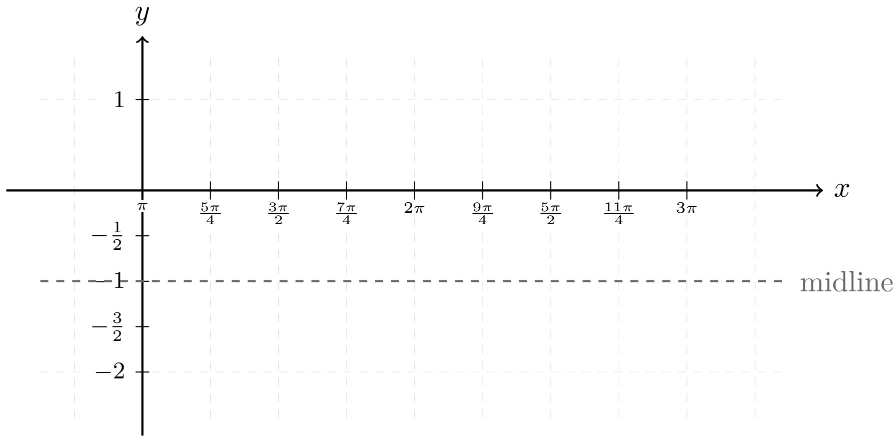
Example 2

Sketch the graph of $y = -\sin\left(x + \dfrac{\pi}{2}\right) + 1$ over two full periods.



Example 3

Sketch the graph of $y = \frac{1}{2}\cos(2x - 2\pi) - 1$ over two full periods.



5.3 Mathematical Modeling with Sine and Cosine

Objectives

- Identify real-world cyclic applications that can be modeled using trigonometric functions.
- Calculate the parameters a, b, c, d of a sinusoidal model directly from data.
- Solve trigonometric equations and inequalities to find safe windows of time in real-world contexts.
- Convert between sine and cosine models for the same dataset.

Building Sinusoidal Models

Many real-world phenomena exhibit periodic behavior (e.g., temperatures, tides, Ferris wheels). We can model these using:

$$y = a\cos\big(b(t-c)\big) + d \qquad \text{or} \qquad y = a\sin\big(b(t-c)\big) + d$$

To find the parameters from a set of data:

1. **Midline / Vertical Shift (d):** The average of the maximum and minimum values:

$$d = \underline{\hspace{3em}}$$

2. **Amplitude (a):** Half the difference between the maximum and minimum values:

$$a = \underline{\hspace{3em}}$$

3. **Period (P) and Frequency (b):** The period is twice the horizontal distance between a consecutive maximum and minimum. Then, compute b:

$$P = \underline{\hspace{3em}} \quad \longrightarrow \quad b = \underline{\hspace{3em}}$$

4. **Phase Shift (c):**
   - For a **cosine model** (assuming $a > 0$), c is the time where a ________ occurs.
   - For a **sine model** (assuming $a > 0$), c is the time where the function crosses the midline going ________.

Tidal Depth Modeling Example

Example Problem Setup

Throughout the day, the depth of the water at the end of a dock varies with the tides. The table shows the depths y (in feet) at various times during the morning, where $t = 0$ represents midnight.

Time	Midnight	2 a.m.	4 a.m.	6 a.m.	8 a.m.	10 a.m.	Noon
Hours since midnight (t)	0	2	4	6	8	10	12
Depth (y, feet)	3.4	8.7	11.3 (Max)	9.1	3.8	0.1 (Min)	1.2

Example 1 (Part A)

Use a **cosine function** of the form $y = a\cos\bigl(b(t-c)\bigr) + d$ to model this data.

Example 1 (Part B)

A boat needs at least **10 feet** of water to moor at the dock. During what times in the **afternoon** (between noon $t = 12$ and midnight $t = 24$) can it safely dock? Round to the nearest minute.

Example 1 (Part C)

Find an equivalent **sine model** of the form $y = a\sin(b(t-c)) + d$ for this data.

5.4 Graphs of Other Trigonometric Functions

Objectives

- Graph the parent tangent, cotangent, secant, and cosecant functions.
- Identify key characteristics including period, domain, range, asymptotes, and symmetry for all six trigonometric functions.
- Sketch transformed tangent and cotangent graphs using vertical stretch, period, phase shift, and vertical translation.
- Sketch transformed secant and cosecant graphs using reciprocal guide curves (sine and cosine).

Tangent and Cotangent parent features

- **Tangent:** $f(x) = \tan x$
  - **Period:** ________ **Domain:** ________
  - **Vertical Asymptotes:** ________ **Range:** ______________
  - x**-intercepts:** ________ **Parity:** ________ (origin symmetry)
- **Cotangent:** $f(x) = \cot x$
  - **Period:** ________ **Domain:** ________
  - **Vertical Asymptotes:** ________ **Range:** ______________
  - x**-intercepts:** ________ **Parity:** ________ (origin symmetry)

where $n \in \mathbb{Z}$. Note that their period is π, unlike sine and cosine which have a period of 2π.

Secant and Cosecant parent features

- **Cosecant:** $f(x) = \csc x = \frac{1}{\sin x}$
  - **Period:** ________ **Domain:** ________
  - **Vertical Asymptotes:** ________ **Range:** ______________
  - x**-intercepts:** ________ **Parity:** ________ (origin symmetry)
- **Secant:** $f(x) = \sec x = \frac{1}{\cos x}$
  - **Period:** ________ **Domain:** ________
  - **Vertical Asymptotes:** ________ **Range:** ______________
  - x**-intercepts:** ________ **Parity:** ________ (y-axis symmetry)

Parent T-Tables (5 Key Features)

$y = \tan x$

x	$y = \tan x$
$-\pi/2$	undefined (VA)
$-\pi/4$	-1
0	0
$\pi/4$	1
$\pi/2$	undefined (VA)

$y = \cot x$

x	$y = \cot x$
0	undefined (VA)
$\pi/4$	1
$\pi/2$	0
$3\pi/4$	-1
π	undefined (VA)

Transformed Tangent and Cotangent Graphs

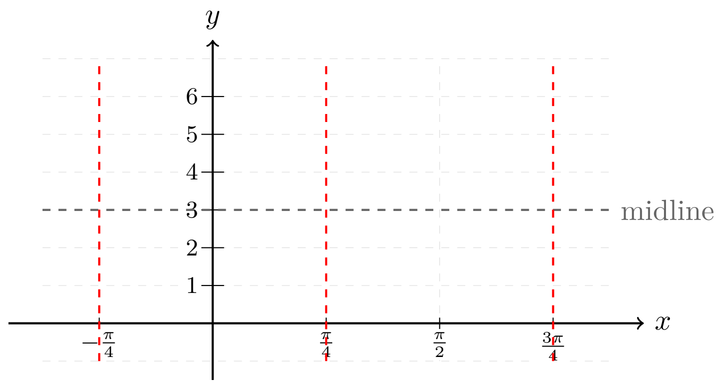
Fact

For $y = a\tan\big(b(x-c)\big) + d$ and $y = a\cot\big(b(x-c)\big) + d$:

- **Period:** ________ (uses π, not 2π).
- **Vertical Stretch/Compress:** Controlled by a. If $a < 0$, the graph is reflected vertically.
- **Phase Shift:** c (horizontal translation).
- **Vertical Translation:** d (midline is $y = d$).

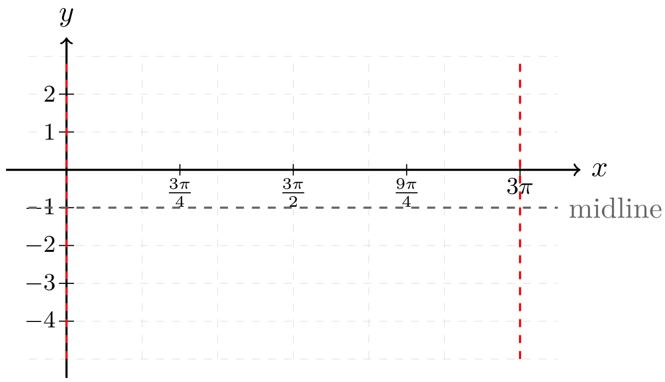
Example 1

Sketch two periods of $y = \tan(2x) + 3$.



Example 2

Sketch the graph of $y = 2\cot\left(\frac{x}{3}\right) - 1$ over one period.



Transformed Secant and Cosecant Graphs

Fact

The Reciprocal Graphing Strategy: Since $\csc\theta = \frac{1}{\sin\theta}$ and $\sec\theta = \frac{1}{\cos\theta}$, we sketch transformed secant/cosecant graphs using their reciprocal guide curves:

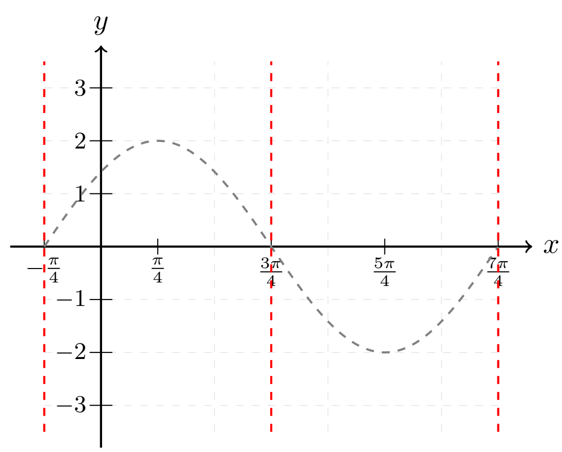
1. **Guide Curve:** Lightly sketch (dashed) the corresponding sin or cos curve with the same parameters:

$$y_{\text{guide}} = a\sin\big(b(x-c)\big) + d \qquad \text{or} \qquad y_{\text{guide}} = a\cos\big(b(x-c)\big) + d$$

2. **Vertical Asymptotes:** Wherever the guide curve crosses its **midline** ($y = d$), the reciprocal function is undefined and has a ________ ________.
3. **Vertices (Key Points):** Wherever the guide curve reaches a local maximum or minimum, the reciprocal curve touches the guide curve and opens in the opposite direction.
   - A guide curve maximum becomes a local minimum (turning up).
   - A guide curve minimum becomes a local maximum (turning down).

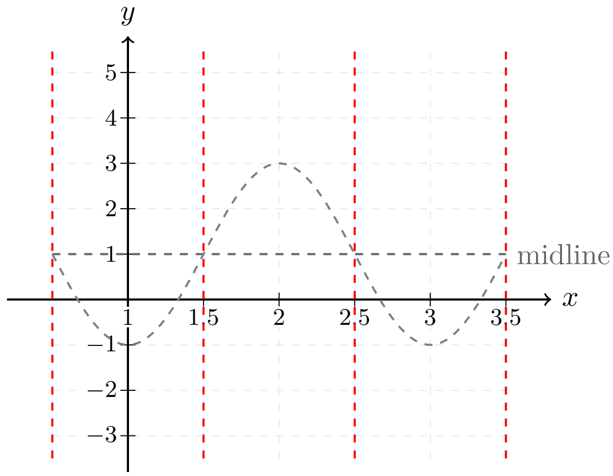
Example 3

Sketch the graph of $y = 2\csc\left(x + \dfrac{\pi}{4}\right)$ over one period.



Example 4

Sketch the graph of $y = -2\sec(\pi x - \pi) + 1$ over one period.



5.5 Inverse Trigonometric Functions

Objectives

- Understand the domain restrictions required to define inverse trigonometric functions.
- Evaluate inverse trigonometric functions for standard exact values.
- Use a calculator to approximate inverse trigonometric values, including reciprocal functions.
- Evaluate compositions of trigonometric functions inside-out, recognizing cancellation restrictions.
- Use right triangles to evaluate trigonometric compositions involving algebraic expressions.

Inverse Trig Function Concept

A trigonometric function takes an ________ as input and returns a ________ of sides. An inverse trigonometric function takes a ________ as input and returns an ________ as output:

$$f(\text{angle}) = \text{ratio} \quad \Longleftrightarrow \quad f^{-1}(\text{ratio}) = \text{angle}$$

Because trigonometric functions are periodic, they fail the Horizontal Line Test (they are not one-to-one). To define their inverse functions, we must restrict their domains.

The Three Primary Inverse Trigonometric Functions

Inverse Sine

The **inverse sine** function (also written $\arcsin x$ or $\sin^{-1} x$) is defined by restricting the domain of $y = \sin x$ to $[-\frac{\pi}{2}, \frac{\pi}{2}]$.

- **Domain:** ________
- **Range:** ________ (Quadrant I and Quadrant IV, written as negative angles)

Inverse Cosine

The **inverse cosine** function (also written $\arccos x$ or $\cos^{-1} x$) is defined by restricting the domain of $y = \cos x$ to $[0, \pi]$.

- **Domain:** ________

- **Range:** ________ (Quadrant I and Quadrant II)

Inverse Tangent

The **inverse tangent** function (also written $\arctan x$ or $\tan^{-1} x$) is defined by restricting the domain of $y = \tan x$ to $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$.

- **Domain:** ________
- **Range:** ________ (Quadrant I and Quadrant IV, excluding endpoints)

Honors: Domain Transformations of Inverse Trig Functions

Consider the function $y = \arcsin(2x)$. Since the input to arcsin must be in $[-1, 1]$, we solve the inequality:

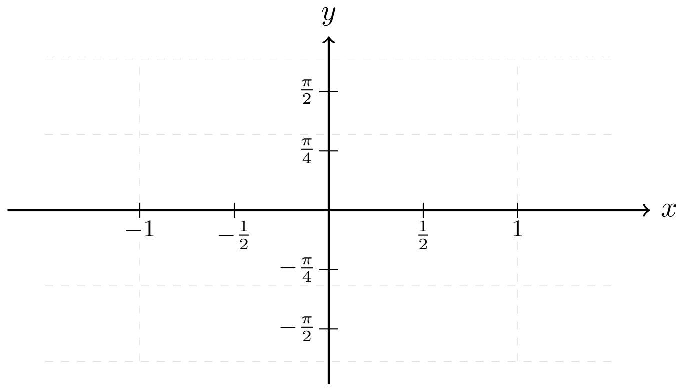
$$-1 \leq 2x \leq 1 \quad \Longrightarrow \quad -\frac{1}{2} \leq x \leq \frac{1}{2}$$

Thus, the domain of $y = \arcsin(2x)$ is compressed to $\left[-\frac{1}{2}, \frac{1}{2}\right]$.

Examples: Evaluating Exact Values

Example 1

Sketch the parent graph of $y = \arcsin(x)$ by hand. Label endpoints.



Example 2–3

Find the exact value of each expression without a calculator (if possible).

2) $\arcsin\left(-\frac{1}{2}\right)$

3) $\sin^{-1}(2)$

Example 4–6

Use a calculator to approximate each value to four decimal places.

4) $\arctan(-8.45)$

5) $\arccos(2)$

6) $\sec^{-1}(3.4)$

Compositions of Trigonometric Functions

Fact

Inverse Function Properties (Cancellation Rules):

- $\sin(\arcsin x) = x$ **only** if $x \in$ ________
- $\arcsin(\sin x) = x$ **only** if $x \in$ ________
- $\cos(\arccos x) = x$ **only** if $x \in$ ________
- $\arccos(\cos x) = x$ **only** if $x \in$ ________

If the input angle x lies outside the restricted range, we must evaluate the expression **inside-out**.

Example 7–8

Find the exact value of each composition (if possible).

7) $\arcsin\left(\sin\dfrac{5\pi}{3}\right)$

8) $\tan[\arctan(-5)]$

Compositions Involving Right Triangles

Example 9–10

Find the exact value or write as an algebraic expression using right triangle sketch methods.

9) $\tan\left(\arccos\dfrac{2}{3}\right)$

10) $\cot(\arccos(3x))$, where $0 \leq x \leq \frac{1}{3}$